

Demonstration of Proposed Features

Date	03 July 2026
Team ID	
Project Name	Predicting Human Development Index (HDI) Using Machine Learning
Maximum Marks	1 Mark

1. Overview

This document guides stakeholders through the core features of the Human Development Index (HDI) Prediction System, demonstrating its capabilities, design, and usability.

2. Demonstrated Features

2.1 Feature 1: Landing Page Dashboard (home.html)

Visuals:

A responsive layout utilizing a dark gradient background with floating particle effects.

Components:

- Hero section explaining the goal of the HDI Predictor.
- Three glassmorphic cards representing the three dimensions of HDI: Health, Knowledge/Education, and Standard of Living.
- Call-To-Action (CTA) button leading users to the interactive prediction form.

2.2 Feature 2: Interactive Input Form (indexnew.html)

Visuals:

A 2-column validated form accepting the key developmental indicators as inputs.

Form Inputs & Ranges:

- Country name (categorical dropdown loaded from training encoder).
- Life Expectancy (50 to 85 years).
- Expected Years of Schooling (5 to 20 years).
- Mean Years of Schooling (3 to 15 years).
- GNI per Capita (\$500 to \$75,000).

2.3 Feature 3: Dynamic Result Visualizations

Visuals:

Real-time prediction rendering featuring score badges, gauges, and insights.

Key Components:

- Predicted Score: Large numerical indicator.
- Tier Badge: Standard UNDP categories (Very High, High, Medium, Low) color-coded appropriately (green, blue, orange, red).
- HDI Gauge: A gradient progress meter positioning the country's level on the full human development spectrum.
- Actionable Insights: Targeted recommendations for improvement based on the predicted tier.

3. Feature Demonstrators & Presenters

To ensure a structured demonstration, each proposed feature of the HDI Prediction System will be showcased by its respective builder/developer during the presentation session:

Proposed Feature / Module	Primary Presenter	Assigned Role
Feature 1: Landing Page Dashboard (home.html)	Jaswanth Yadav Keerthi Sai Kiran Bandi	Web Developer / UI Designer
Feature 2: Interactive Input Form (indexnew.html)	Jaswanth Yadav Keerthi Swapna Beldhari	Web Developer / UI Designer
Feature 3: Dynamic Result Visualizations (Gauge & Insights)	Swapna Beldhari Jaswanth Yadav Keerthi	Data Scientist / ML Engineer
Session Overview & Joint Q&A Segment	Sai Kiran Bandi	Lead Developer / Moderator

4. Demonstration of Proposed Features:

This document presents the features proposed during the planning phase of the HDI Prediction System project and confirms whether each feature was successfully implemented and demonstrated. The project was developed to provide an intelligent Machine Learning-based HDI prediction system that assists policymakers in assessing development levels using socio-economic indicators.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	HDI Score Prediction	Predicts continuous HDI scores based on country and socio-economic indicators (Life Expectancy, Schooling, GNI).	IMPLEMENTED	YES	Successfully demonstrated using the trained Linear Regression model.
2	Machine Learning Model Integration	Flask application loads the trained Linear Regression model and LabelEncoder to generate predictions.	IMPLEMENTED	YES	Model loaded successfully and produced accurate predictions.
3	Input Validation	Validates all socio-economic inputs on both client-side and server-side before prediction.	IMPLEMENTED	YES	Prevents invalid or incomplete input values.
4	Prediction Result Dashboard	Displays predicted HDI score, development tier (Very High, High, Medium, Low), and policy recommendations.	IMPLEMENTED	YES	Prediction results displayed dynamically after user input.
5	Data Visualization Dashboard	Generates statistical charts including strip plots, heatmaps, and actual vs. predicted graphs.	IMPLEMENTED	YES	Visualizations successfully generated during model analysis.
6	Responsive Web Interface	Provides a responsive and user-friendly interface accessible from desktop and mobile browsers.	IMPLEMENTED	YES	Successfully demonstrated across different screen sizes.
7	Prediction History	Maintains recent prediction history	IMPLEMENTED	YES	Enables users to review previous

		during the active user session.			prediction results.
8	Cloud Deployment Support	Project is structured for future deployment on IBM Cloud, Render, or similar cloud platforms.	PARTIAL	NO	Architecture supports deployment; production deployment is planned as future work.

5. Feature Implementation Summary:

Metric	Value
Total Features Proposed	8
Total Features Implemented	7
Total Features Demonstrated	7
Overall Implementation Rate	87.5%